

Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 4: Following the Pollen Tube: How Double Fertilisation Explains Why a Lone Pawpaw Flowers but Sets No Fruit

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	D	C	B	D	B	C

USE AFTER: PREDICT PHASE — Predict: written and drawn prediction of the route of the male cell, using the lone pawpaw and cassava cutting sketch

Q1. A pollen grain lands on the stigma of a compatible flower. Which route do the male nuclei take to reach the egg cell?

- A. The pollen grain mixes with nectar and flows down the stem to the ovule
- B. The pollen grain falls straight through the petals onto the egg cell
- C. The pollen grain stays on the stigma and the egg cell moves up the style to meet it
- D. The pollen grain germinates and a pollen tube grows down the style to the ovary, then into the ovule to the egg cell**

Correct answer: D. After landing on the stigma the pollen grain germinates, and the pollen tube carries the male nuclei down the style to the ovary and ovule. Pollen never lands directly on the egg cell.

USE AFTER: OBSERVE PHASE — Observe: pathway diagram worksheet, animation paused at micropyle entry

Q2. A pollen tube has grown down the style and into the ovary and an ovule. Through which structure does it enter the ovule to reach the embryo sac?

- A. The filament
- B. The petal base
- C. The stigma
- D. The micropyle**

Correct answer: D. The pollen tube enters the ovule through a small opening called the micropyle and then releases the sperm nuclei into the embryo sac. The stigma is where pollen lands, at the start of the route.

USE AFTER: OBSERVE PHASE — Observe: animation paused at the two fusions of double fertilisation, and the evidence table

Q3. When a pollen tube reaches the embryo sac, it releases two sperm nuclei. What does each sperm nucleus do?

- A. One sperm nucleus fuses with the tube nucleus, and the other fuses with the egg cell
- B. One sperm nucleus fuses with the egg cell, and the other disappears because it is not needed
- C. One sperm nucleus fuses with the egg cell, and the other fuses with the two polar nuclei**
- D. Both sperm nuclei fuse with the egg cell to make a strong zygote

Correct answer: C. Double fertilisation involves two fusions. One sperm nucleus joins the egg cell to form the zygote. The other joins the two polar nuclei to form the endosperm nucleus.

USE AFTER: EXPLAIN PHASE — Explain: Think-Pair-Share on double fertilisation, correcting the misconception that pollination is fertilisation

Q4. A learner says: "Pollination and fertilisation are the same thing." Which statement correctly corrects this learner?

- A. Pollination happens inside the ovule, and fertilisation happens on the stigma
- B. Pollination is the transfer of pollen to the stigma, and fertilisation is the fusion of the sperm nuclei with the egg cell and polar nuclei**
- C. Pollination is the fusion of a sperm nucleus with an egg cell, and fertilisation is the landing of pollen on the stigma
- D. Pollination happens only in maize, and fertilisation happens only in beans

Correct answer: B. Pollination is only the transfer of pollen to the stigma. Fertilisation is the later fusion of nuclei inside the embryo sac, after the pollen tube has delivered the sperm nuclei.

USE AFTER: EXPLAIN PHASE — Explain: individual Claim-Evidence-Reasoning paragraph on why a lone pawpaw or avocado tree flowers but sets no fruit

Q5. A single pawpaw tree in a compound in Kisumu is covered in flowers but sets no fruit. There is no other pawpaw tree nearby. Which explanation is best supported by what happens inside a flower?

- A. The tree has too many flowers, so the pollen is used up before it reaches the stigma
- B. The tree makes seeds without fertilisation, but they fall off before they can be seen
- C. The flowers need to be cut off so that the stem can grow into a new plant
- D. No compatible pollen arrives, so no pollen tube grows, no gametes fuse, and no zygote or endosperm forms, so there is no seed or fruit**

Correct answer: D. Fruit and seed formation needs fertilisation. Without compatible pollen there is no pollen tube and no fusion, so no zygote or endosperm forms and the flowers do not set fruit.

USE AFTER: MODEL BUILDING PHASE — Model: pair oral probe 'Why is the endosperm triploid?' and the revised class poster of products of double fertilisation

Q6. In maize, the egg cell, each sperm nucleus and each polar nucleus have 10 chromosomes. One sperm nucleus fuses with the two polar nuclei to form the endosperm nucleus. How many chromosomes does the endosperm nucleus have?

- A. 40
- B. 30**
- C. 20
- D. 10

Correct answer: B. The endosperm nucleus is formed from one sperm nucleus and two polar nuclei, which is three nuclei of 10 chromosomes each. $3 \times 10 = 30$, so the endosperm is triploid. The value 20 is the number in the zygote, which forms from the other fusion.

USE AFTER: END OF LESSON — End of lesson: revised class model comparing the Asexual branch (cassava cuttings) with the Sexual branch (maize seeds)

Q7. A Kenyan farmer grows cassava from stem cuttings and maize from seeds. Which statement correctly contrasts the two methods?

- A. Cassava cuttings need a compatible partner plant for pollen, while maize seeds do not
- B. Maize grains form without fertilisation, while cassava cuttings form a zygote and an endosperm
- C. Cassava cuttings grow into new plants without any fusion of gametes, while a maize grain forms only after fertilisation of the egg cell and polar nuclei**
- D. Both cassava cuttings and maize grains form only after the fusion of gametes

Correct answer: C. A cassava cutting is asexual propagation, so no gametes fuse and the new plant is identical to the parent. A maize grain, with its embryo and endosperm, forms only after double fertilisation.